

Mobile ERP System

**Stop losing margin to
disconnected marketplaces.**

A full-stack operational tool built for modern commerce.

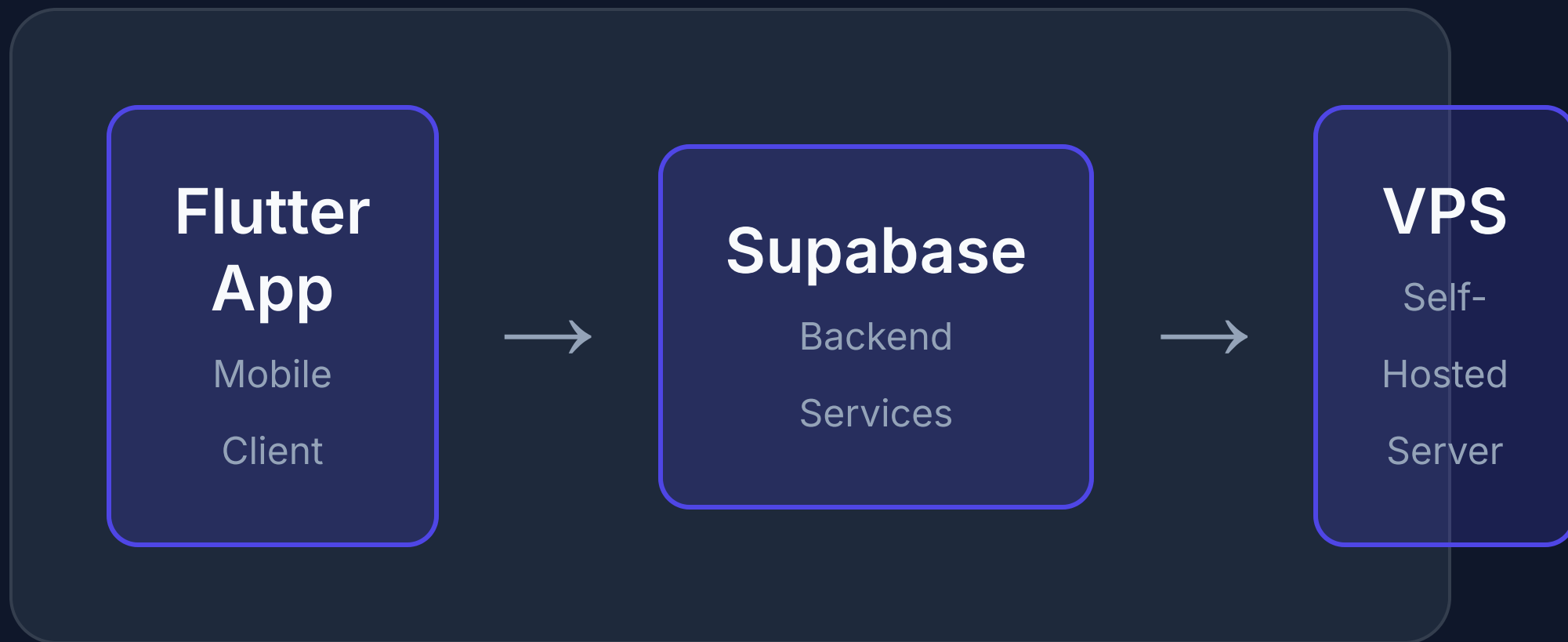
Portfolio Project

The Problem

- **Inventory Syncing** across multiple marketplaces (Shopee, TikTok) is manual and error-prone.
- **Finance teams** struggle to match marketplace payouts with actual COGS.
- **Warehouse staff** lack a unified mobile interface for daily operations.

What I Built

A unified ERP bridging the gap between physical stock and digital sales channels.



Architecture, Simplified

- **Client:** Cross-platform Flutter App
- **Core Services:** Supabase (Auth, Postgres, Storage, Realtime)
- **External APIs:** Shopee & TikTok Shop Integrations
- **Infrastructure:** Dockerized environment on a dedicated VPS

Choosing to self-host Supabase on a VPS wasn't just about reducing costs—it was about **control**.

- Total data ownership and privacy.
- Bypassed expensive managed-BaaS scaling limits.
- Provided hands-on experience with Docker, server hardening, and DevOps workflows.

Omnichannel Sync

Live integration with major e-commerce platforms.

Shopee & TikTok APIs

Automated webhooks keep local inventory and online stock in perfect sync, preventing stockouts and overselling.

Finance Reconciliation

Ending the spreadsheet nightmare for accounting teams.

- Cross-checks incoming marketplace payouts against expected revenue.
- Calculates true profit by incorporating precise COGS (Cost of Goods Sold).
- Highlights discrepancies and missing funds instantly.

Role-Based Workflows

Built for the whole team, with security baked in.

Warehouse Staff

Focused on picking, packing, and stock adjustments.

Finance Team

Access to sensitive COGS and revenue reconciliation data.

Production readiness goes beyond functional code.

- **Security:** Strict Row Level Security (RLS) policies in Postgres ensuring tenant isolation.
- **Infrastructure:** Hardened VPS with firewall rules and secure SSH practices.
- **Reliability:** Automated database backups and edge function monitoring.

Key takeaways

Lessons learned shipping a complex full-stack app.

- Balancing mobile UX constraints with data-heavy ERP requirements.
- Managing complex database migrations safely in production.
- Handling eventual consistency across third-party marketplace APIs.

Skills Demonstrated

A proven track record across the stack.

Flutter & Dart

PostgreSQL Design

Infrastructure & VPS

Docker

API Integration

Role-Based Security

Data Reconciliation

Let's Connect

Open to IT Support & IT Specialist
roles.

Ready to bring full-stack problem solving
to your operational challenges.

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